

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Winter 2022

Course: TY. B.Tech. Branch: AI&DS. Semester: VI

Subject Code & Name: BTAIOE604A & Big Data Analytics.

Max Marks: 60

Date: 19-07-2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) What is Big Data? Explain types of Digital Data? (BT2/CO1) 6
- B) Explain Five V's of Big Data? (BT2/CO1) 6
- C) What is Big Data Analysis? Explain three real time application of Big Data? (BT2/CO1) 6

Q.2 Solve Any Two of the following.

- A) Draw Hadoop Eco-System? Explain in short Hadoop Ecosystem? (BT4/CO2) 6
- B) Explain HDFS architecture with Name Node, Data Node and Secondary Node with Diagram? (BT2/CO2) 6
- C) Differentiate between Flume and Scoop? (BT4/CO2) 6

Q.3 Solve Any Two of the following.

- A) What is mean by Job Scheduling? Explain its Types. (BT2/CO3) 6
- B) List features of Map-Reduce? Explain any two Features? (BT1/CO3) 6
- C) Explain in short Shuffle and Sort processing using Map-Reduce process with text "This is an Apple. Apple is red in color ." (BT2/CO3) 6

Q.4 Solve Any Two of the following.

- A) Briefly explain the architecture of Pig? (BT3/CO4) 6
- B) Differentiate between RDBMS and HBase? (BT4/CO4) 6
- C) Illustrate HiveQL data definition language and explain any two Hive QL DDL command with its syntax and example? (BT2/CO4) 6

Q.5 Solve Any Two of the following.

- A) Explain concept of Kafka with its architecture? (BT2/CO5) 6
- B) Describe features and components of Apache Spark? (BT2/CO5) 6
- C) Elaborate Components of Kerberos authentication with Diagram? (BT2/CO5) 6

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Regular 2023

Course: TY. B.Tech. Branch: Artificial Intelligence and Data Science.

Semester :V Subject Code & Name: BTAIPE603D Web Development

Max Marks: 60

Date:17/07/23

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- | | | |
|---|-----------|---|
| A) What is www? Explain HTTP Request and Response message. | (BT1/CO1) | 6 |
| B) Differentiate between HTML & HTML 5. | (BT3/CO1) | 6 |
| C) What are the advantages of CSS? Explain types of CSS. | (BT2/CO1) | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|--|-----------|---|
| A) Describe about Java script. Write 6 features of it. | (BT2/CO2) | 6 |
| B) Explain advantage and Enlist disadvantage of JSP over the Servlet. | (BT4/CO2) | 6 |
| C) Write short note on DOM. Explain any 3 event handling in DOM. | (BT3/CO2) | 6 |

Q.3 Solve Any Two of the following.

- | | | |
|--|-----------|---|
| A) What is servlet? Explain life cycle of servlet along with diagram. | (BT1/CO3) | 6 |
| B) What is XML? Explain how it is different from HTML. Write benefit of it. | (BT3/CO3) | 6 |
| C) Describe AJAX. Explain working of it with diagram. | (BT4/CO3) | 6 |

Q.4 Solve Any Two of the following.

- | | | |
|---|-----------|---|
| A) Differentiate JSP & Servlet. | (BT2/CO4) | 6 |
| B) What is web services? Write 6 advantages of it. | (BT2/CO4) | 6 |
| C) What is struts? Explain MVC in struts with suitable diagram. | (BT3/CO4) | 6 |

Q.5 Solve Any Two of the following.

- | | | |
|---|-----------|---|
| A) What is PHP? Explain its data type & Array in PHP. | (BT2/CO5) | 6 |
| B) What are the advantages of ASP.NET? Explain client-side and server side validation in web pages. | (BT2/CO5) | 6 |
| C) Explain .NET framework along with suitable diagram. Write Short note on Node JS. | (BT3/CO5) | 6 |

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examination – 2023

Course: B. Tech.

Branch : AI&ML / AIDS

Semester : VI

Subject Code & Name: BTAICE602 , Advanced Machine Learning

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain why Principal Component Analysis (PCA) is used for Dimensionality Reduction?	L2 / CO1	6
B) How to determine K using the Silhouette Method & TWSS Method?	L3 / CO1	6
C) What is the main objective of clustering algorithms?	L3 / CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain the concepts of support and confidence in association rule mining?	L1 / CO2	6
B) What is the Apriori algorithm, and how does it work?	L3 / CO2	6
C) What is collaborative filtering in recommendation systems?	L2 / CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain the concepts of agent, environment, and rewards in reinforcement learning?	L2 / CO3	6
B) What is the role of exploration and exploitation in reinforcement learning?	L3 / CO3	6
C) Explain the concept of the Q-learning algorithm?	L1 / CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain the concepts of trend, seasonality, and noise in time series data?	L1 / CO4	6
B) What is the autocorrelation function (ACF) in time series analysis?	L2 / CO4	6
C) Briefly explain the concept of forecasting in time series analysis?	L2 / CO4	6
Q. 5 Solve Any Two of the following.		12
A) What is model selection in machine learning, and why is it important?	L3 / CO5	6
B) Discuss the main advantages and disadvantages of boosting algorithms?	L1 / CO5	6
C) Explain the concept of cross-validation and how it helps in model selection?	L1 / CO5	6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2023

Course: B. Tech

Branch: AI & DS

Semester: VI

Subject Code & Name: BTAIHM605B/BTCOE605B

Employability and Skill Development

Max Marks: 60

Date: 21/07/2023

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as OBE or the Course Outcome (CO) on which question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculator is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1. Solve any two of the following.		12
A) What is the distinction between soft skills and hard skills, and how do they contribute to professional success?	CO1	6
B) Write a job application letter for the post of Junior Engineer. (Attach Chronological Resume)	CO1	6
C) How does interdisciplinary collaboration contribute in resolving complex industrial problems?	CO1	6
Q.2. Solve any two of the following.		12
A) Define mental arithmetic skills and describe its importance in our life.	CO2	6
B) Explain in detail the Multiple Intelligence Theory.	CO2	6
C) Describe the types of higher order thinking skills using Bloom's Taxonomy.	CO2	6
Q.3. Solve any two of the following.		12
A) Write a descriptive paragraph on National Education Policy – 2020. (Paragraph of 150-200 words)	CO3	6
B) Define story writing and describe the essential components that contribute to the creation of a captivating and well-crafted story.	CO3	6
C) Write an acceptance email to Infosys Limited which has sent you an offer letter for the post of System Analyst based on your interview.	CO3	6

Q. 4. Solve any two of the following. 12

- ~~A)~~ Explain the types and key principles of Group Discussion. CO4 6
- B) How non-verbal communication plays significant role in Interview situation. CO4 6
- ~~C)~~ What are the fundamental principles of active listening? CO4 6

Q. 5. Solve any two of the following. 12

- A) Describe problem solving model in detail. CO5 6
- ~~B)~~ Write a brief note on importance of communication at workplace. CO5 6
- ~~C)~~ Explain the importance of Brainstorming in problem solving. CO5 6

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examination – 2023

Course: B. Tech.

Branch: CSE(AI&ML)

Semester: VI

Subject Code & Name: BTAIC601 / Deep Learning

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
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	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Differentiate between Artificial Neural Network and Biological Neural Network.	(L4 / CO1)	6
B) Explain perceptron learning for ANN.	(L1 / CO1)	6
C) Compare and contrast linearly separable and linearly non separable problems with suitable example.	(L2 / CO1)	6
Q.2 Solve Any Two of the following.		12
A) Describe the architecture of Convolutional Neural Network.	(L1 / CO2)	6
B) In CNN what are the pros and cons of using max pooling layer Vs. Average pooling layer.	(L2 / CO2)	6
C) Enlist and justify the advantages of using Convolutional Neural Network Vs. Dense Neural Network for image classification.	(L5 / CO2)	6
Q. 3 Solve Any Two of the following.		12
A) What are the different ways of transfer learning? Explain them in brief.	(L1 / CO3)	6
B) Explain back propagation learning algorithm in detail.	(L1 / CO3)	6
C) Discuss the different techniques for handling overfitting issues during deep learning process.	(L2 / CO3)	6
Q.4 Solve Any Two of the following.		12
A) What is a Recurrent Neural Network (RNN)?	(L1 / CO4)	6
B) What is "Dead Neuron Problem"? How can Leaky ReLU overcome this problem?	(L2 / CO4)	6
C) Discuss in detail the implementation of AND gate using Mc Culloch Pitts neuron model.	(L2 / CO4)	6

Q.5 Solve Any Two of the following.

12

A) Explain and enlist the applications of Neural Style Transfer. –

(L1 / CO5) 6

B) How image denoising is achieved using Deep Learning techniques.

(L4 / CO5) 6

C) Write Short note on Generative Adversarial Network.

(L2 / CO5) 6

*** End ***

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